

On Smoothly Varying Frame Timing in First-Person Gaming

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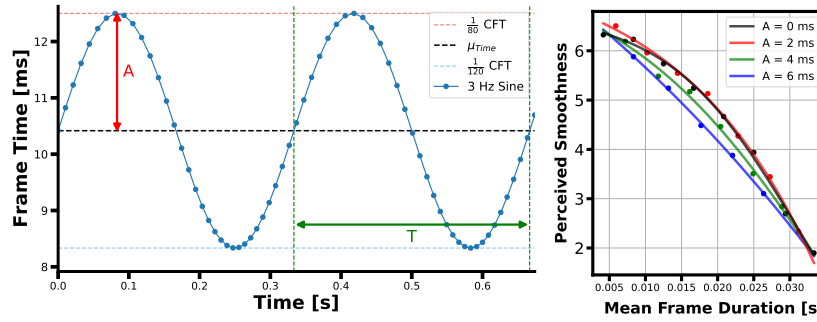


Figure 1: (Left) Two periods of a smoothly varying, sinusoidal frame time with 2 ms amplitude (A), 333.33 ms period (T), and time-weighted mean frame time depicted as a dashed black line ($\mu_{Time} = 10.42$ ms). (Right) A plot of a user rating of "smoothness" under varying mean frame time and amplitude (2-6 ms). Generally smoothness decreases with increased frame time, as expected, with additional degradations of smoothness coming from variable frame time (VFT).

CCS CONCEPTS

• Human-centered computing → Displays and imagers; Pointing; User studies; • Applied computing → Computer games.

KEYWORDS

frame rate, frame time, first-person targeting, first-person games, perception, performance

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1 INTRODUCTION

In video games, the time between rendered frames varies significantly and is often uncontrolled. With the advent of variable refresh rate (VRR) displays, like G-SYNC and FreeSync, players can experience varying frame timing as produced instead of aliasing to nearby refresh timings (e.g. VSYNC On). We investigate how the presentation of smoothly varying frame times over time (e.g., Fig. 1, Left) affects a gamer's perception of smoothness and aiming performance. Participants performed a first person targeting task

while experiencing sinusoidally varying frame times of different amplitude and mean duration. We found a decrease in perceptual smoothness ratings in the presence of varying frame times.

2 BACKGROUND

Some researchers have used G-Sync as a part of stimulus design [Poth et al. 2018], or to promote energy savings [Ko 2019; You et al. 2020]. Others have used VRR as a method for shifting refresh rates to match a perceptual goal, but neglected to study the impact of varying the refresh rate itself [Denes et al. 2020; Jindal et al. 2021]. More directly related works study VRR's effect on task performance, some showing positive general outcomes [Riahi and Watson 2021; Watson et al. 2019], while others have focused on the perceptual effects (quality of experience) when artificially adding variability [Liu et al. 2023]. None of these studies carefully control the frame time sequence and distribution, nor study rates above 120 Hz. Our study fills these gaps using a smoothly varying controlled sequence, including stimuli up to 240 Hz. We also study player performance measures on atomic FPS tasks to complement our analysis of the perceptual measures, similar to previous studies [Claypool and Claypool 2007; Shim et al. 2011; Spjut et al. 2019a] with a focus on target elimination time as our metric of interest.

3 EXPERIMENT DESIGN

We conducted two experiments using an FPS experimentation platform, First Person Science [Boudaoud et al. 2022; Spjut et al. 2019b] (release v22.05.01), to collect data from participants [Klein et al. 2023, 2024]. In particular, we use a feature whereby an experimenter can specify a target frame time to be used on a per-frame basis.

The first experiment displays a group of moving green targets on screen for four seconds, during which the user attempts to click on

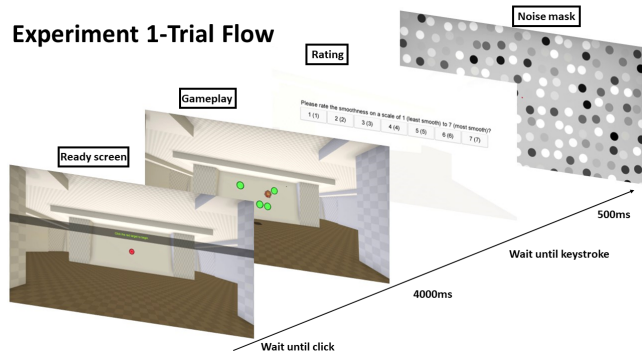


Figure 2: Trial flow for experiment 1. Note that the visual noise mask is presented in the final stage of the flow.

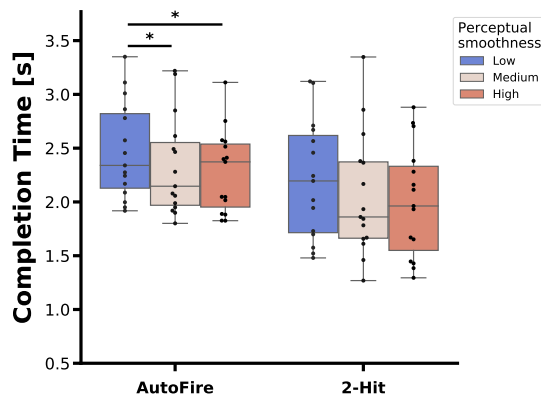


Figure 3: Time to destroy the target in Experiment 2. Box plots with individual data points per subject ($n = 15$) superimposed on top. Horizontal black bars with the * symbol indicate statistically significant comparisons.

the targets while experiencing the determined frame time pattern selected in a semi-random balanced order. After each trial, the user rates smoothness on a Likert scale ranging from 1, least smooth to 7, smoothest. Then a short visual noise mask is shown before the next trial begins. This flow is summarized visually in Fig. 2. In the second experiment, in order to assess aiming performance, the user destroys a green target by tracking it with the mouse (auto fire) or clicking it twice (2-hit). No smoothness survey results were collected with this task. The conditions were grouped in sets of similar smoothness ratings as determined by experiment 1.

4 RESULTS

The results from the first experiment (Fig. 1, right) demonstrate a monotonic relationship between perceptual smoothness ratings and mean frame duration, as expected. However, there is an additional minor effect from increasing the sine wave amplitude above 2 ms, with higher amplitudes leading to lower perceived smoothness around a common mean frame duration. Task completion time results from the second experiment only reached statistical significance between different smoothness ratings in auto fire. No statistical significance was found for task completion time beyond the controlled smoothness rating in auto fire (Fig. 3).

5 CONCLUSIONS

In this work, we studied the impact of smoothly varying frame timing on player perception and performance. While there are clear signs of perceptual smoothness being reduced by frame timing variability (experiment 1), when controlling for perceptual smoothness in experiment 2 we saw no additional impact on player performance in the 2-hit and tracking tasks. Though these experiments are intended to be representative of first person shooter game play, they may not fully generalize to all types of gaming experiences under the same variation in frame timing. That said, perceptual smoothness corresponds to aiming performance in FPS games. In particular, when using variable refresh rate displays, users can trust their feeling of smoothness as a soft proxy for aiming performance.

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